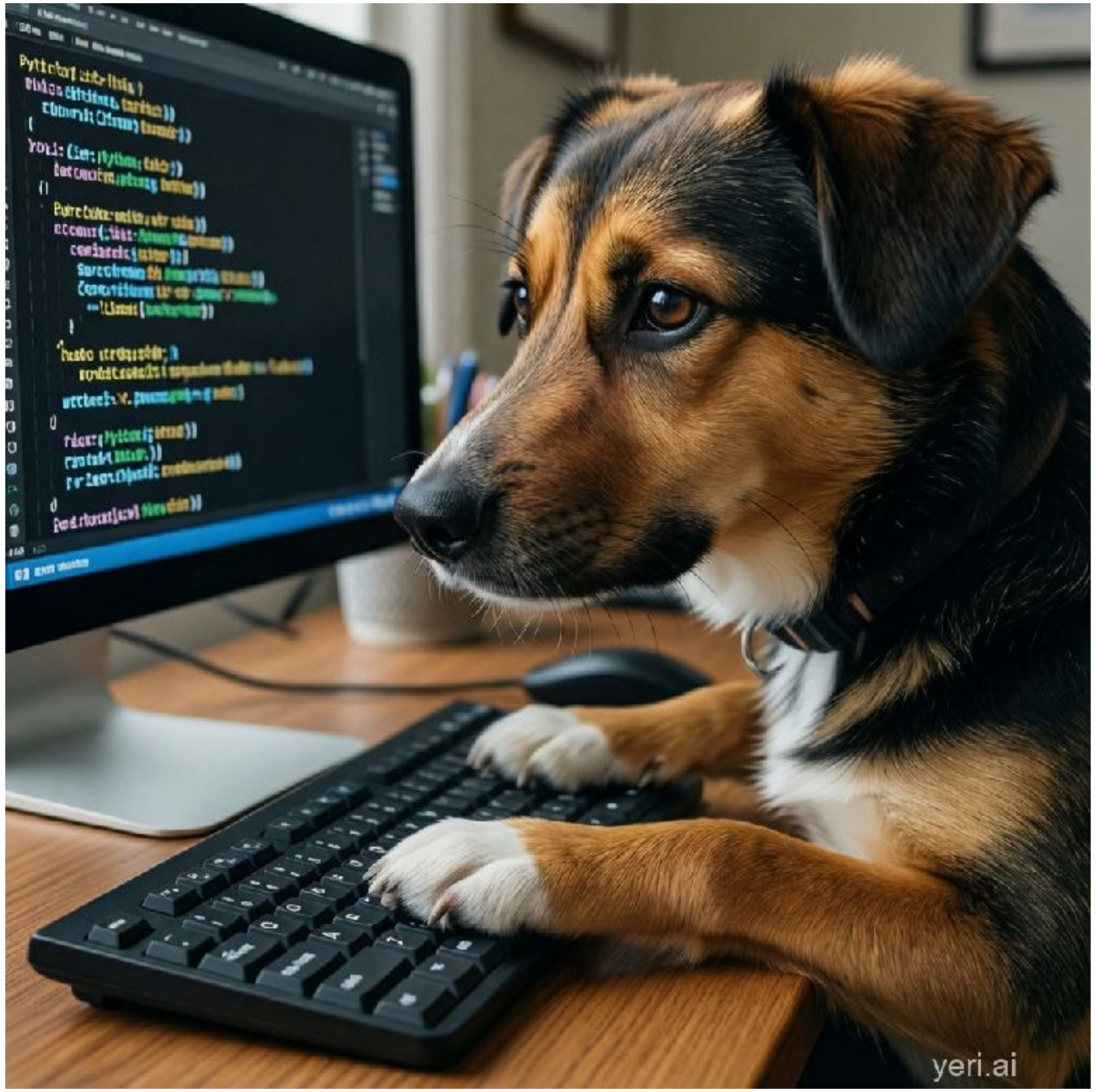
Diffusion Model

Yiding Chen

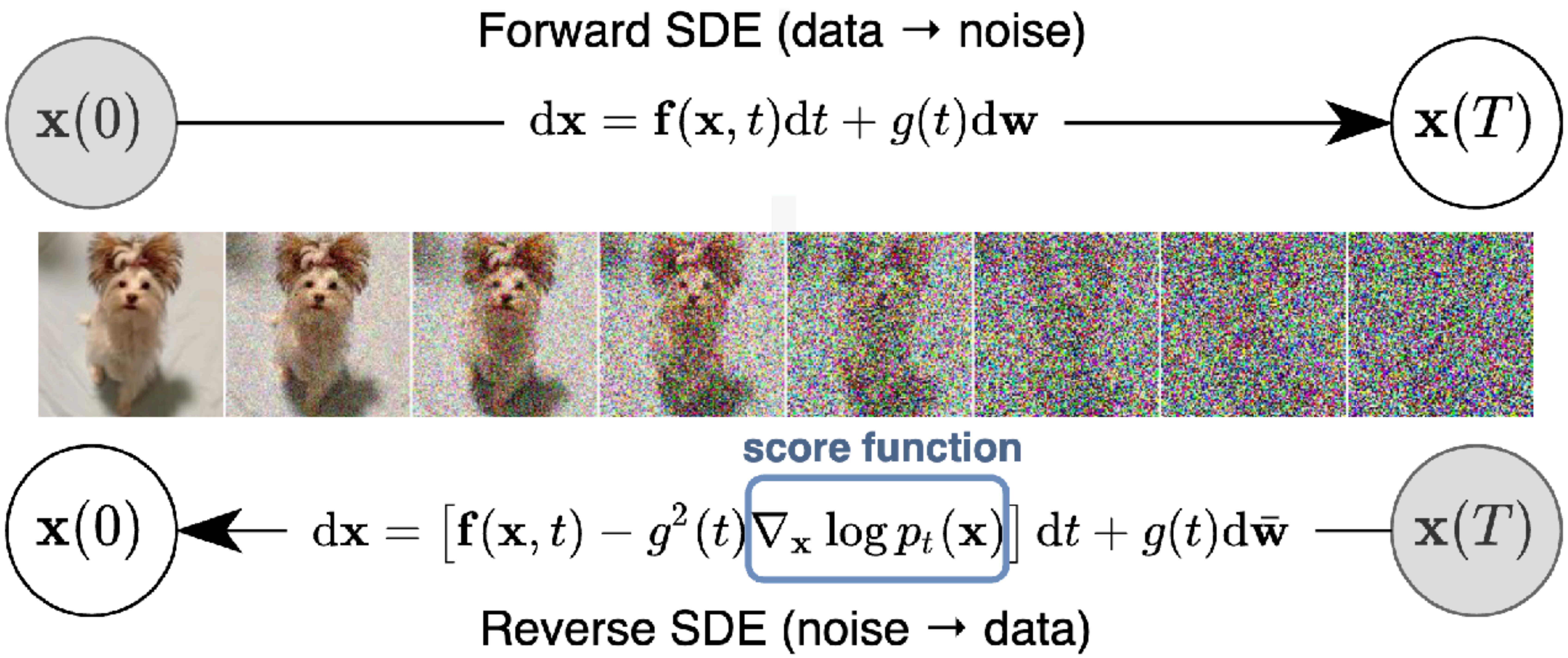
Prompt

A dog typing Python code on a keyboard, ultra-realistic style.



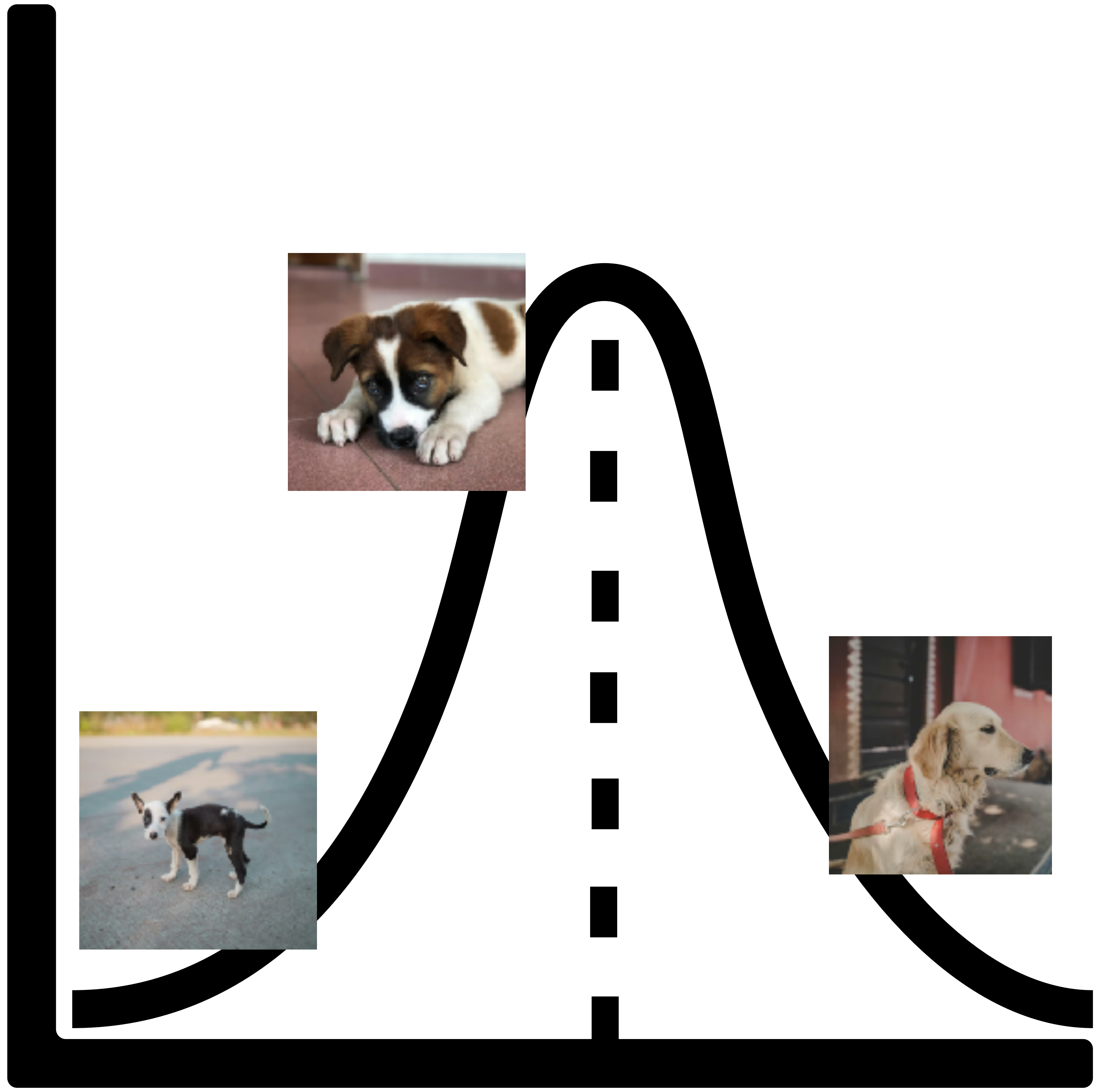
1. Diffusion Model Basics
2. Efficient Diffusion Sampling
3. Applications Beyond Image Generation

Diffusion model



The sampling problem

Generate **sample** x $=$ 

from **distribution** P_{data} $=$ 

Forward process: Ornstein–Uhlenbeck process

$$\mathrm{d}\, x_t = \boxed{-x_t \mathrm{d}\, t} + \boxed{\sqrt{2}\mathrm{d}\, w_t} \qquad x_0 \sim P_{\text{data}}$$

OU process is the continuous time limit of

$$x_{t+\Delta t} = \underbrace{\boxed{x_t - x_t \cdot \Delta t}}_{\text{Shrink to zero}} + \underbrace{\boxed{\sqrt{2} \cdot \sqrt{\Delta t} \cdot Z_t}}_{\text{Add noise}} \qquad Z_t \sim \mathcal{N}(0, I)$$

Limiting Behavior of the OU Process

$$\mathrm{d}\,x_t = -\,x_t \mathrm{d}\,t + \sqrt{2}\mathrm{d}\,w_t \qquad x_0 \sim P_{\text{data}}$$

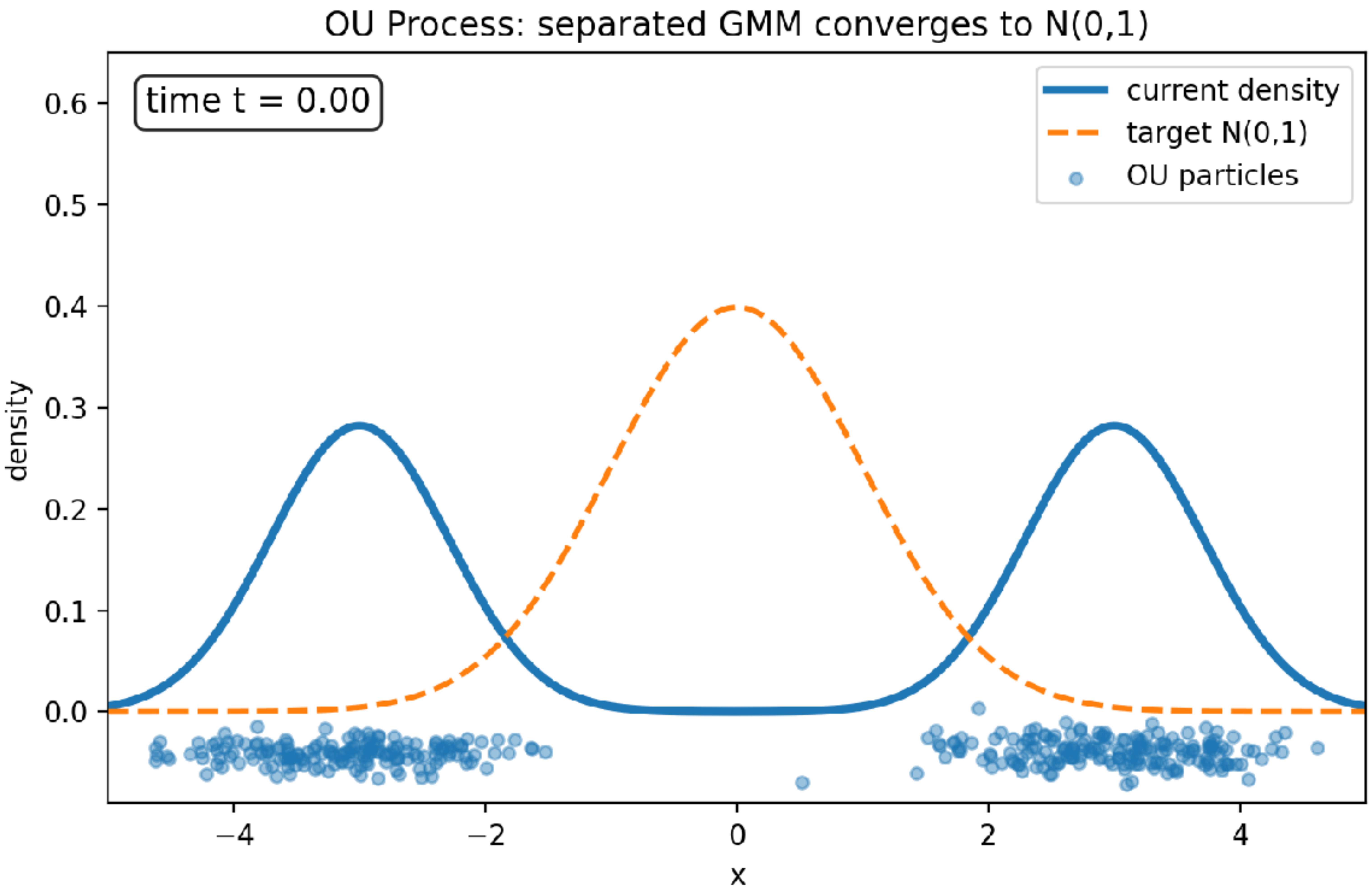
Conditional distribution

$$x_t \,|\, x_0 \sim \mathcal{N}(e^{-t}x_0, (1 - e^{-2t})I)$$

For a given x_0, when $t \;\rightarrow\; \infty$,

$$\mathcal{N}(e^{-t}x_0, (1 - e^{-2t})I) \;\rightarrow\; \mathcal{N}(0, I)$$

OU Process simulation



Reverse Process of the OU Diffusion

OU process:

$$\mathrm{d}\, x_t = - x_t \mathrm{d}\, t + \sqrt{2} \mathrm{d}\, w_t \qquad x_0 \sim P_{\text{data}}$$

Let q_t be the PDF of x_t

The reverse process is:

score function

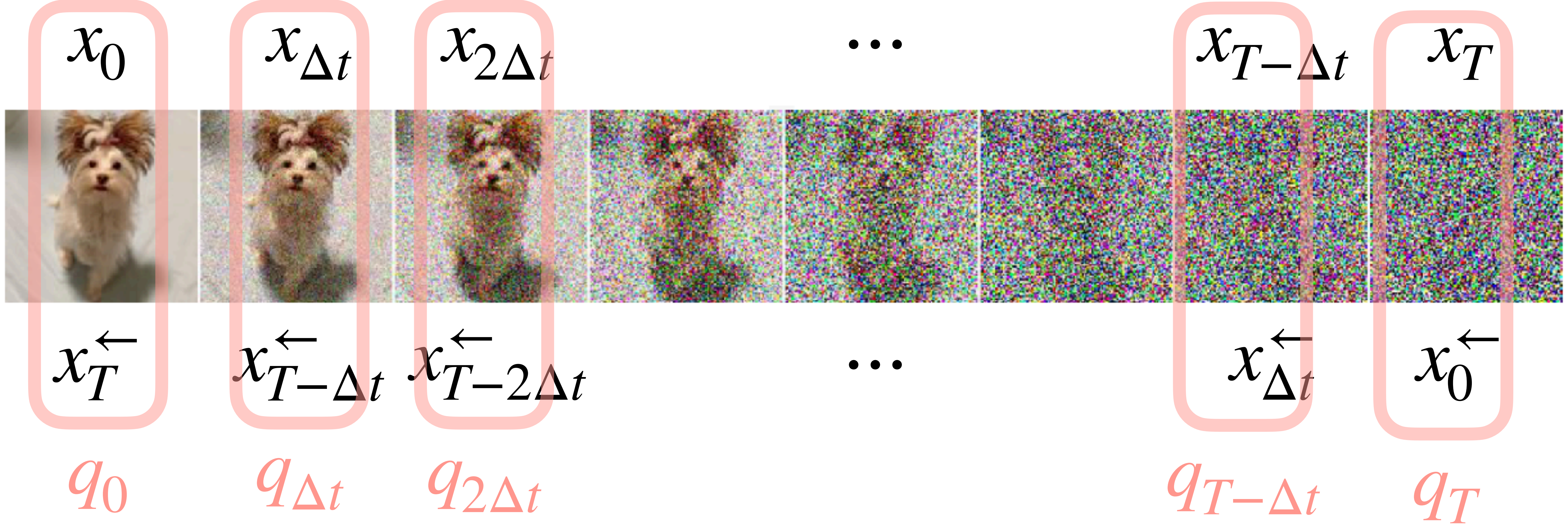
$$\mathrm{d}\, x_t^{\leftarrow} = (x_t^{\leftarrow} + 2 \underbrace{\nabla \ln q_{T-t}(x_t^{\leftarrow})}) \mathrm{d}\, t + \sqrt{2} \mathrm{d}\, w_t \qquad x_0^{\leftarrow} \sim q_T$$

Reverse Diffusion Sampling: Error Sources

- Initial distribution mismatch ($q_T \approx \mathcal{N}(0,I)$) ✅
- Marginal distribution of forward vs marginal distribution of reverse
- Score approximation error
- Discretization error

Forward and Reverse OU Share the Same Marginals

Theorem: q_{T-t} is the PDF of the reverse process.



Proof idea: show that q_{T-t} satisfies the FP equation of the reverse process

1-D Fokker-Planck equation

Suppose x_t follows SDE

$$\mathrm{d}\, x_t = f(x_t, t)\mathrm{d}\, t + \sqrt{2}\mathrm{d}\, w_t$$

Let p_t be the PDF of x_t, then p_t evolves according to:

$$\frac{\partial}{\partial t} p_t(x) = -\frac{\partial}{\partial x}[f(x, t)p_t(x)] + \frac{\partial^2}{\partial x^2} p_t(x)$$

The SDE describes how samples move;

the FP equation describes how their distribution moves.

Proof of the Theorem *Consider 1-D case for simplicity*

FP equation of the reverse process is:

$$\frac{\partial}{\partial t}p_t(x) = -\frac{\partial}{\partial x}[(x + 2\frac{\partial}{\partial x}\ln q_{T-t}(x))p_t(x)] + \frac{\partial^2}{\partial x^2}p_t(x)$$

We will verify q_{T-t} satisfies it.

Proof of the Theorem *Consider 1-D case for simplicity*

Plug in $p_t = q_{T-t}$:

$$\text{RHS} = -\frac{\partial}{\partial x}[(x + 2\frac{\partial}{\partial x}\ln q_{T-t}(x))q_{T-t}(x)] + \frac{\partial^2}{\partial x^2}q_{T-t}(x)$$

$$= -\frac{\partial}{\partial x}[xq_{T-t}(x)] - 2\frac{\partial}{\partial x}[\frac{\partial}{\partial x}q_{T-t}(x))] + \frac{\partial^2}{\partial x^2}q_{T-t}(x)$$

$$= -\frac{\partial}{\partial x}[xq_{T-t}(x)] - \frac{\partial^2}{\partial x^2}q_{T-t}(x)$$

$$= \frac{\partial}{\partial t}q_{T-t}(x) = \text{LHS}$$

$$(\ln f(x))' = \frac{f'(x)}{f(x)}$$

Because q_t satisfies the FP equation of the forward process:

$$\frac{\partial}{\partial t}q_t(x) = \frac{\partial}{\partial x}[xq_t(x)] + \frac{\partial^2}{\partial x^2}q_t(x)$$

Reverse Diffusion Sampling: Error Sources

- Initial distribution mismatch ($q_T \approx \mathcal{N}(0,I)$) ✅
- Marginal distribution of forward vs marginal distribution of reverse ✅
- Score approximation error
- Discretization error

Score function approximation

Score function: $\nabla \ln q_t(\cdot)$

Goal: $\min_\theta \mathbb{E}_{x_t \sim q_t}[\|s_\theta(x_t, t) - \boxed{\nabla \ln q_t(x_t)}\|_2^2]$

unknown

$$\mathbb{E}_{x_t \sim q_t}[\|s_\theta(x_t, t)\|_2^2] - 2\mathbb{E}_{x_t \sim q_t}[\langle s_\theta(x_t, t), \nabla \ln q_t(x_t)\rangle] + \boxed{\mathbb{E}_{x_t \sim q_t}[\|\nabla \ln q_t(x_t)\|_2^2]}$$

constant in θ

Score Matching

$$\mathbb{E}_{x_t \sim q_t}[\langle s_\theta(x_t, t), \nabla \ln q_t(x_t) \rangle] = \int s_\theta(x_t, t)^\top \nabla \ln q_t(x_t) \cdot q_t(x_t) dx_t$$

$$= \int s_\theta(x_t, t)^\top \nabla q_t(x_t) dx_t$$

$$(\ln f(x))' = \frac{f'(x)}{f(x)}$$

$$= \int s_\theta(x_t, t)^\top \nabla \int q(x_t | x_0) q_0(x_0) dx_0 dx_t$$

$$= \int \int s_\theta(x_t, t)^\top \nabla q(x_t | x_0) q_0(x_0) dx_0 dx_t$$

$$= \int \int s_\theta(x_t, t)^\top \nabla \ln q(x_t | x_0) q(x_t | x_0) q_0(x_0) dx_0 dx_t$$

$$= \mathbb{E}_{x_0, x_t | x_0}[\langle s_\theta(x_t, t), \nabla \ln q(x_t | x_0) \rangle]$$

Score Matching

$$\min_\theta \mathbb{E}_{x_t \sim q_t}[\|s_\theta(x_t, t) - \nabla \ln q_t(x_t)\|_2^2]$$

$$\Updownarrow$$

$$\min_\theta \mathbb{E}_{x_t \sim q_t}[\|s_\theta(x_t, t)\|_2^2] - 2\mathbb{E}_{x_0, x_t | x_0}[\langle s_\theta(x_t, t), \nabla \ln q(x_t \mid x_0)\rangle]$$

$$\Updownarrow$$

$$\min_\theta \mathbb{E}_{x_0, x_t | x_0}[\|s_\theta(x_t, t) - \nabla \ln q(x_t \mid x_0)\|_2^2]$$

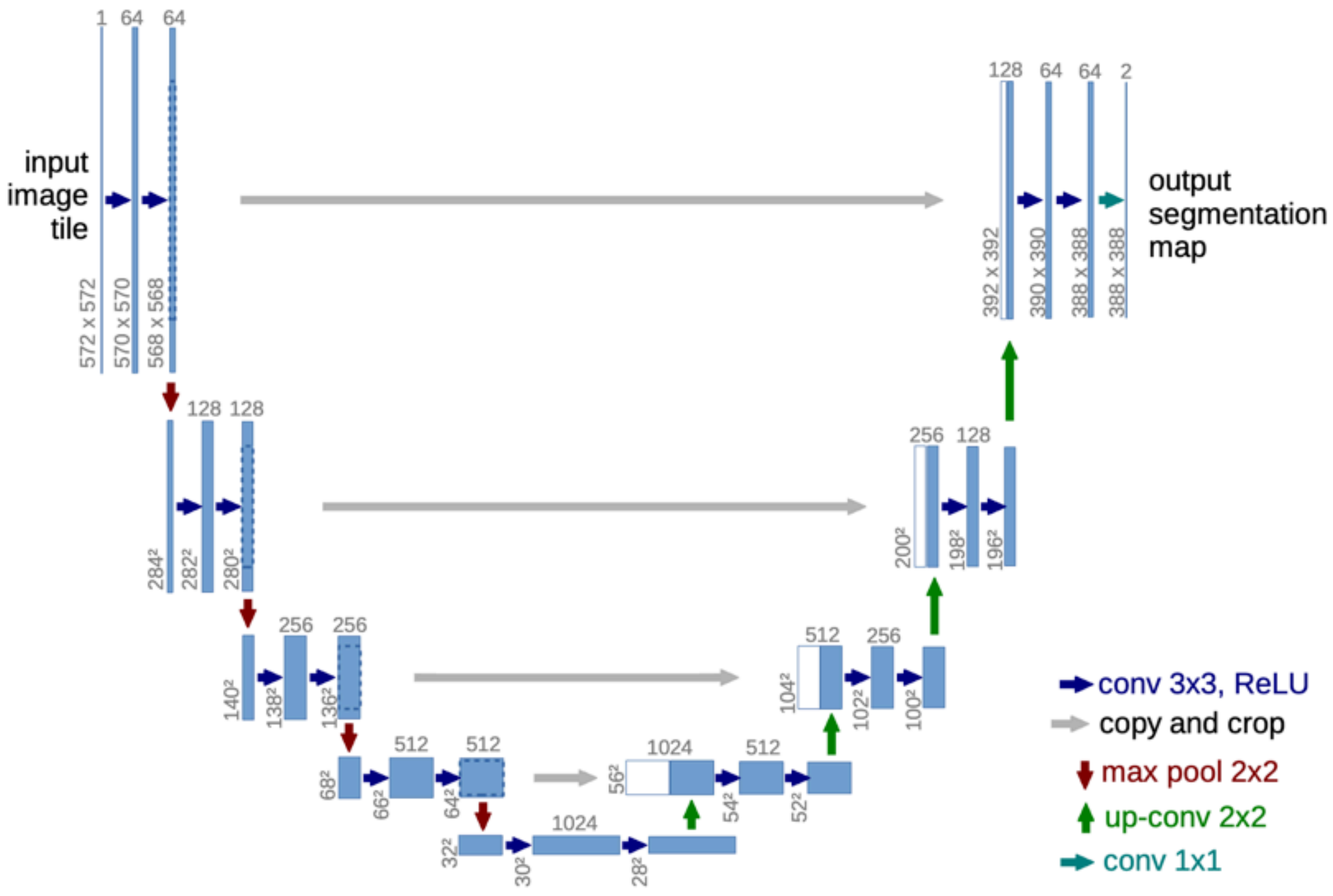
Score Matching

$$\min_{\theta} \mathbb{E}_{x_0, x_t | x_0}[\| s_\theta(x_t, t) - \nabla \ln q(x_t \mid x_0) \|_2^2]$$

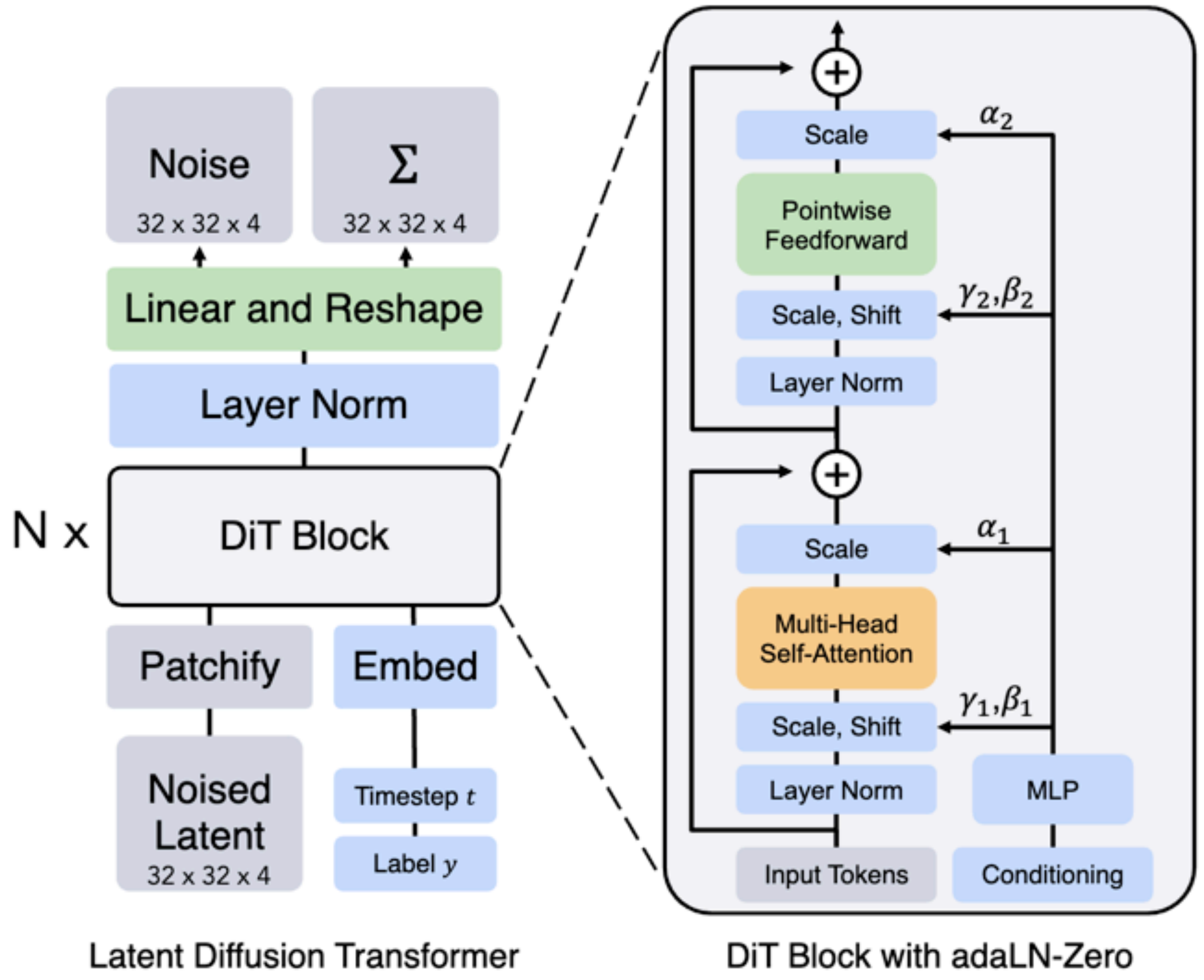
Recall $x_t \mid x_0 \sim \mathcal{N}(e^{-t}x_0, (1 - e^{-2t})I)$, $\nabla \ln q(x_t \mid x_0) = -\dfrac{x_t - e^{-t}x_0}{1 - e^{-2t}}$

Score function is predicting the noise added to x_t

Model architecture



U-Net



Diffusion Transformer

Reverse Diffusion Sampling: Error Sources

- Initial distribution mismatch ($q_T \approx \mathcal{N}(0,I)$) ✅
- Marginal distribution of forward vs marginal distribution of reverse ✅
- Score approximation error ✅
- Discretization error

Diffusion model

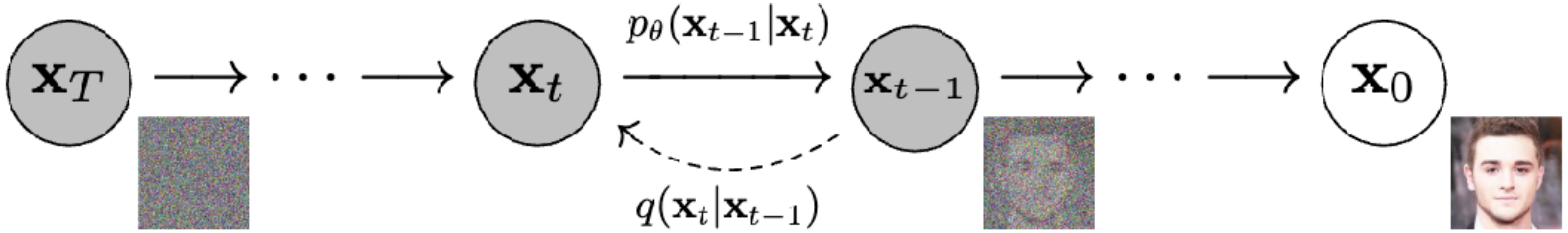
Training:

$$\min_{\theta} \mathbb{E}_{x_0, x_t | x_0}[\|s_\theta(x_t, t) + \frac{x_t - e^{-t}x_0}{1 - e^{-2t}}\|_2^2]$$

Inference:

$$\mathrm{d}\, x_t^{\leftarrow} = (x_t^{\leftarrow} + 2s_\theta(x_t^{\leftarrow}, T - t))\mathrm{d}\, t + \sqrt{2}\mathrm{d}\, w_t \qquad x_0^{\leftarrow} \sim \mathcal{N}(0, I)$$

DDPM



Forward process: $x_t = \sqrt{1-\beta_t}x_{t-1} + \sqrt{\beta_t}Z_t \qquad Z_t \sim \mathcal{N}(0,I)$

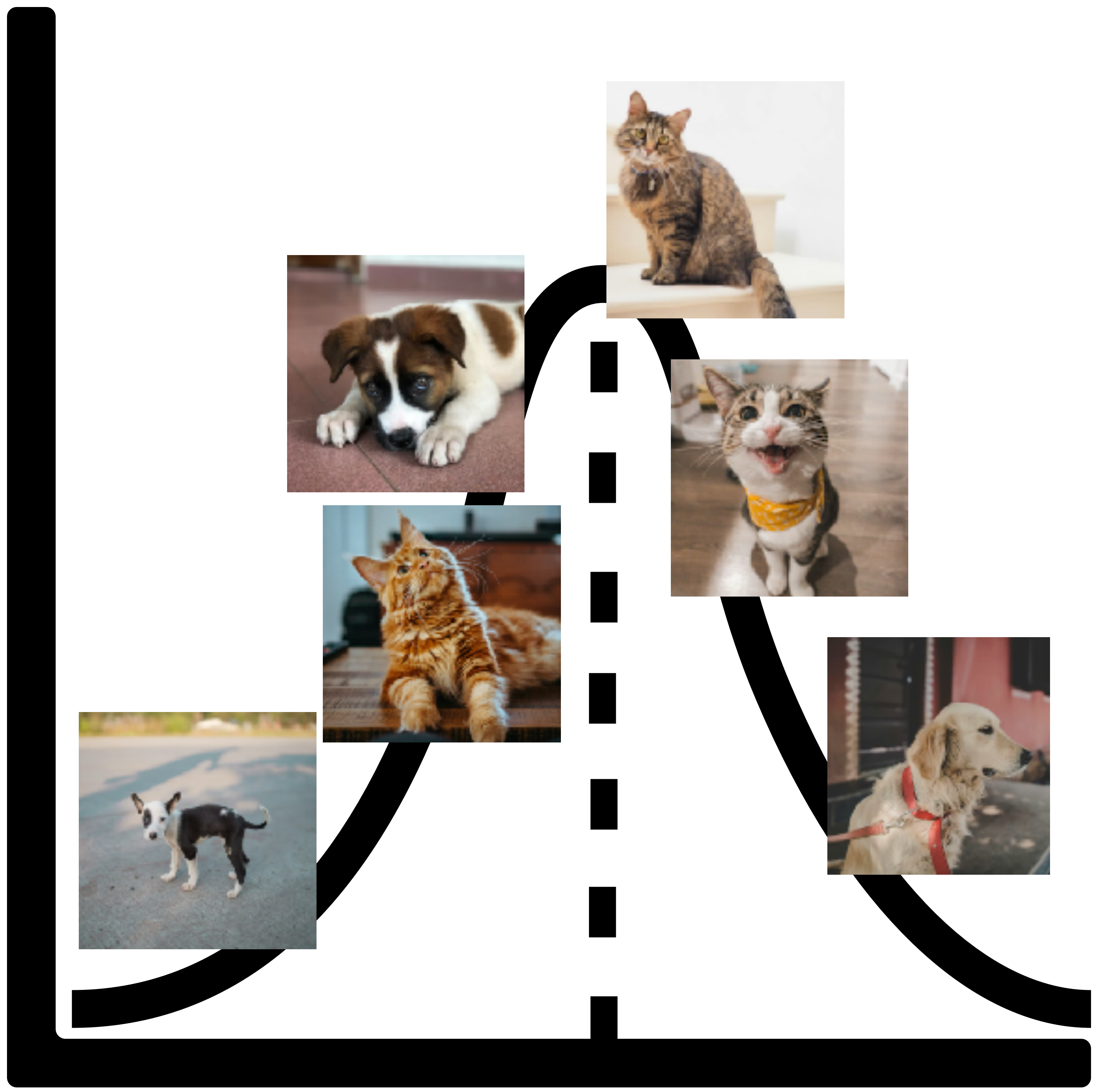
Reverse process: $x_{t-1} = \mu_\theta(x_t, t) + \sigma_t Z_t \qquad Z_t \sim \mathcal{N}(0,I)$

Discrete version of $\mathrm{d}\, x_t = -\frac{1}{2}\beta_t x_t \mathrm{d}\, t + \sqrt{\beta_t}\mathrm{d}\, w_t$

Break

Controlled generation

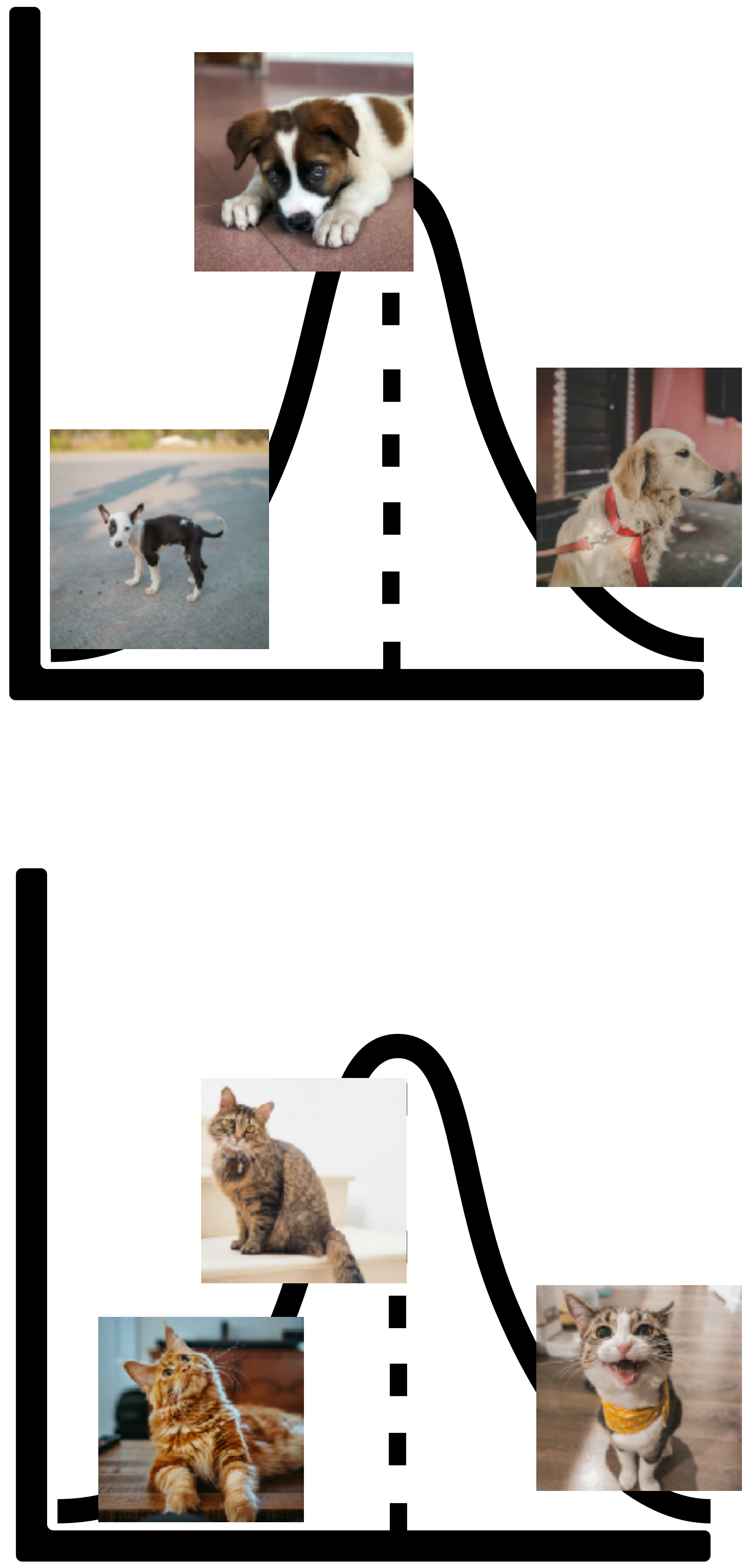
$P(\cdot)$



uncontrolled generation

$P(\cdot \mid \text{Dog})$

$P(\cdot \mid \text{Cat})$



controlled generation

Controlled generation

Goal: generate from $P(x \mid y = i)$ for each i

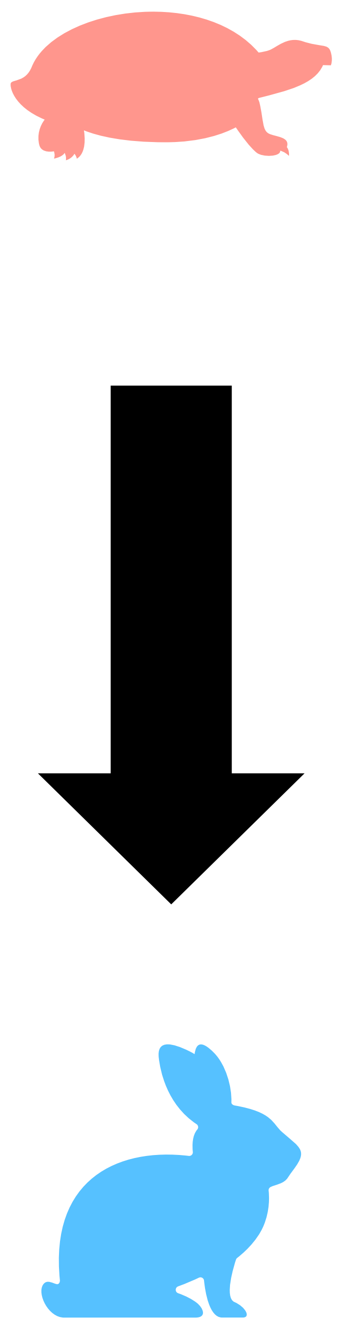
Need: $\nabla \ln p_t(x_t \mid y = i)$ for each i

$$\nabla \ln p_t(x_t \mid y = i) = \nabla \ln \frac{p_t(y = i \mid x_t) \cdot p_t(x_t)}{p(y = i)}$$

$$= \underbrace{\nabla \ln p_t(x_t)}_{\text{Unconditional score function}} + \underbrace{\nabla \ln p_t(y = i \mid x_t)}_{\text{Gradient of classifier}} - \underbrace{\nabla \ln p_t(y = i)}_{\text{zero}}$$

1. Diffusion Model Basics ✅
2. Efficient Diffusion Sampling
3. Applications Beyond Image Generation

Method	**Sampling steps**
DDPM / SDE	~ 1000
DDIM / ODE	20 - 100
Distilled method	< 10



ODE Sampler

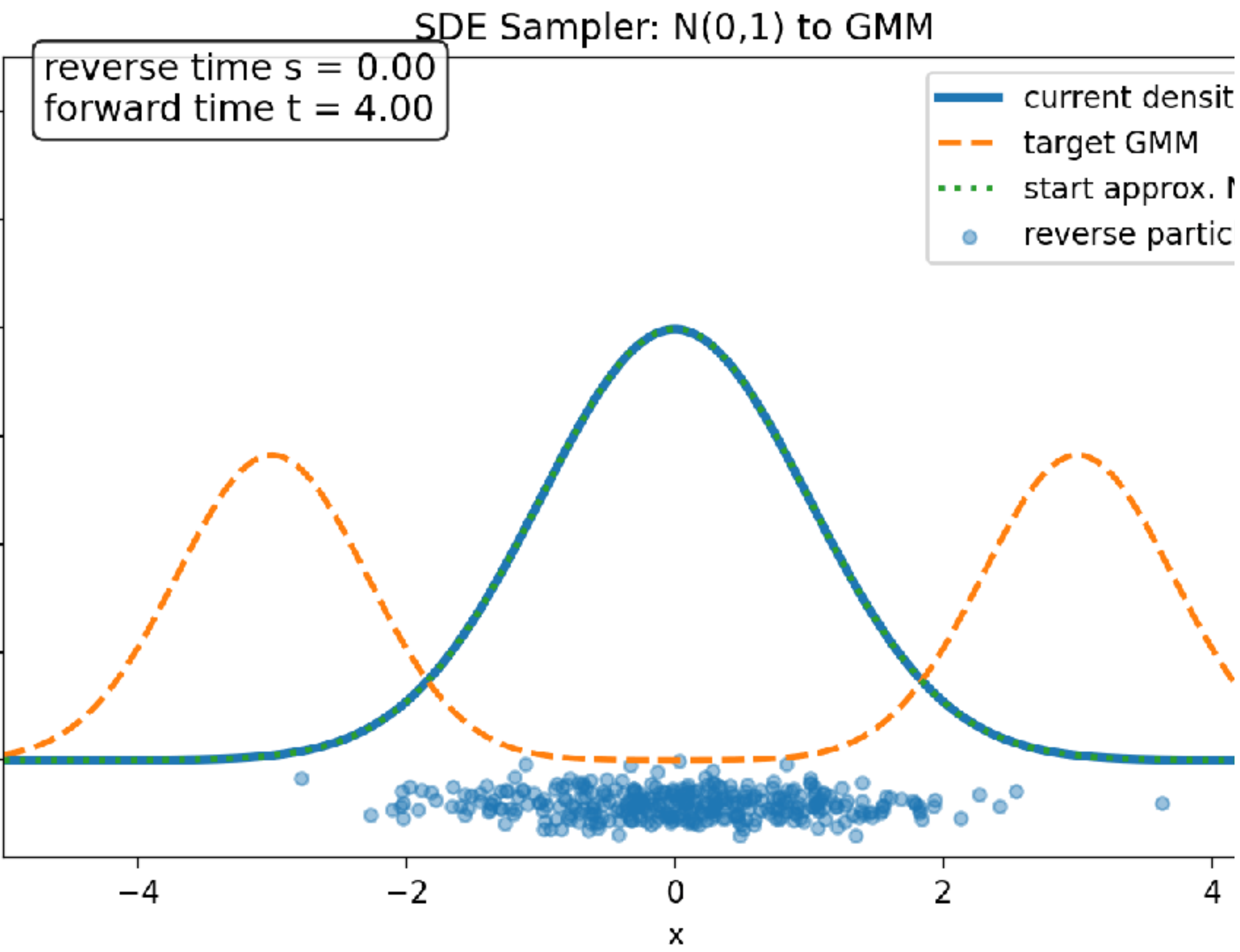
SDE: $\mathrm{d}\, x_t^{\leftarrow} = (x_t^{\leftarrow} + 2\nabla \ln q_{T-t}(x_t^{\leftarrow}))\mathrm{d}\, t + \sqrt{2}\mathrm{d}\, w_t \qquad x_0^{\leftarrow} \sim q_T$

$$\Downarrow$$

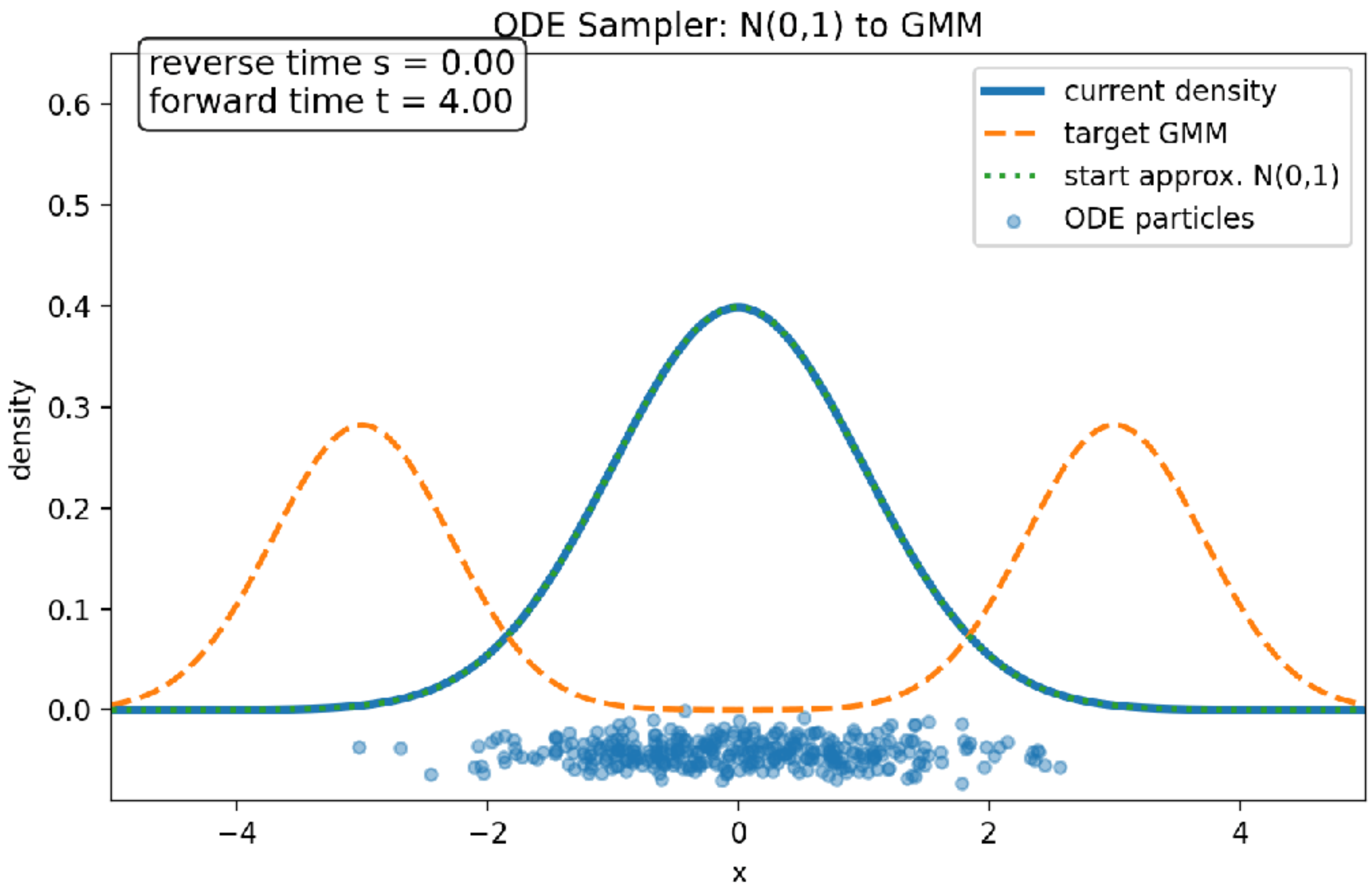
ODE: $\mathrm{d}\, x_t^{\leftarrow} = (x_t^{\leftarrow} + \nabla \ln q_{T-t}(x_t^{\leftarrow}))\mathrm{d}\, t \qquad x_0^{\leftarrow} \sim q_T$

- Same marginal distribution
- Deterministic trajectory with fixed initial
- smoother

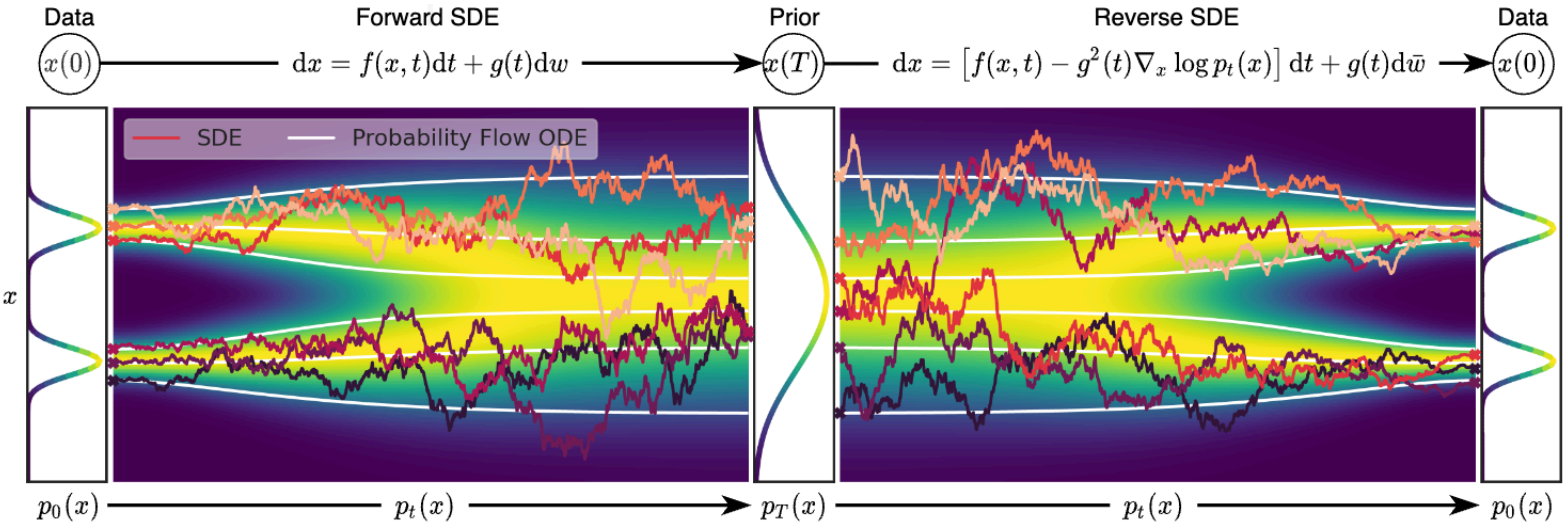
SDE sampler



ODE sampler



ODE Sampler



Data
Forward SDE
Prior
Reverse SDE
Data
$x(0)$
$\mathrm{d}x = f(x,t)\mathrm{d}t + g(t)\mathrm{d}w$
$x(T)$
$\mathrm{d}x = \left[f(x,t) - g^2(t)\nabla_x \log p_t(x)\right]\mathrm{d}t + g(t)\mathrm{d}\bar{w}$
$x(0)$
SDE
Probability Flow ODE
x
$p_0(x)$
$p_t(x)$
$p_T(x)$
$p_t(x)$
$p_0(x)$

ODE sampler

Table 1: CIFAR10 and CelebA image generation measured in FID. $\eta = 1.0$ and $\hat{\sigma}$ are cases of DDPM (although Ho et al. (2020) only considered $T = 1000$ steps, and $S < T$ can be seen as simulating DDPMs trained with S steps), and $\eta = 0.0$ indicates DDIM.

		CIFAR10 (32×32)					CelebA (64×64)				
	S	10	20	50	100	1000	10	20	50	100	1000
ODE	η 0.0	**13.36**	**6.84**	**4.67**	**4.16**	4.04	**17.33**	**13.73**	**9.17**	**6.53**	3.51
	η 0.2	14.04	7.11	4.77	4.25	4.09	17.66	14.11	9.51	6.79	3.64
	η 0.5	16.66	8.35	5.25	4.46	4.29	19.86	16.06	11.01	8.09	4.28
	η 1.0	41.07	18.36	8.01	5.78	4.73	33.12	26.03	18.48	13.93	5.98
SDE	$\hat{\sigma}$	367.43	133.37	32.72	9.99	**3.17**	299.71	183.83	71.71	45.20	**3.26**

Distillation methods

Step of ODE Sampler:

Step 1: Sample $x_0^{\leftarrow} \sim q_T$

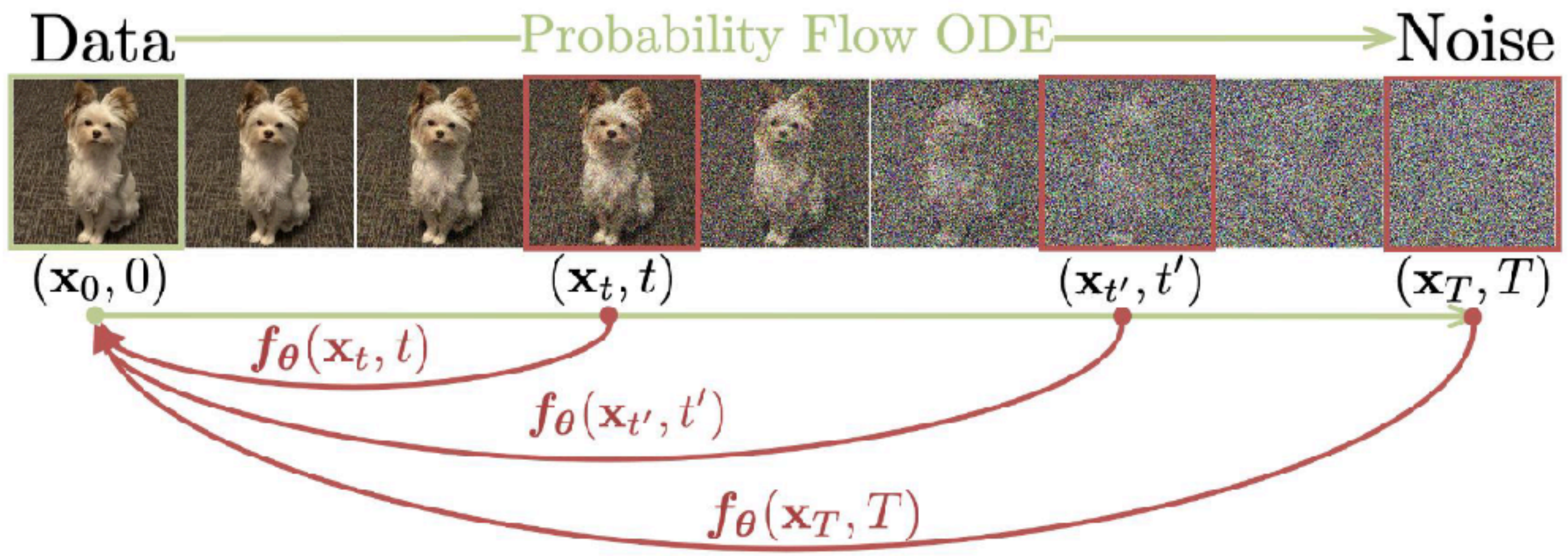
Step 2: Solve $\mathrm{d}\, x_t^{\leftarrow} = (x_t^{\leftarrow} + \nabla \ln q_{T-t}(x_t^{\leftarrow}))\mathrm{d}\, t$

$$x_T^{\leftarrow} \to x_{T-\Delta t}^{\leftarrow} \to x_{T-2\Delta t}^{\leftarrow} \to \cdots \to x_{2\Delta t}^{\leftarrow} \to x_{\Delta t}^{\leftarrow} \to x_0^{\leftarrow}$$

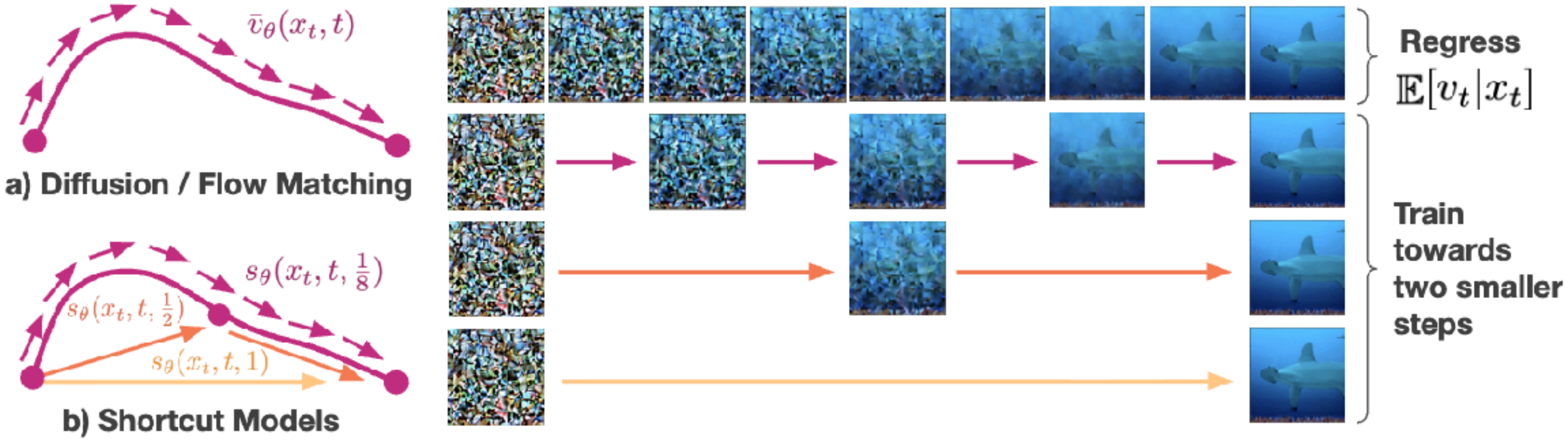
Distill

Distillation methods

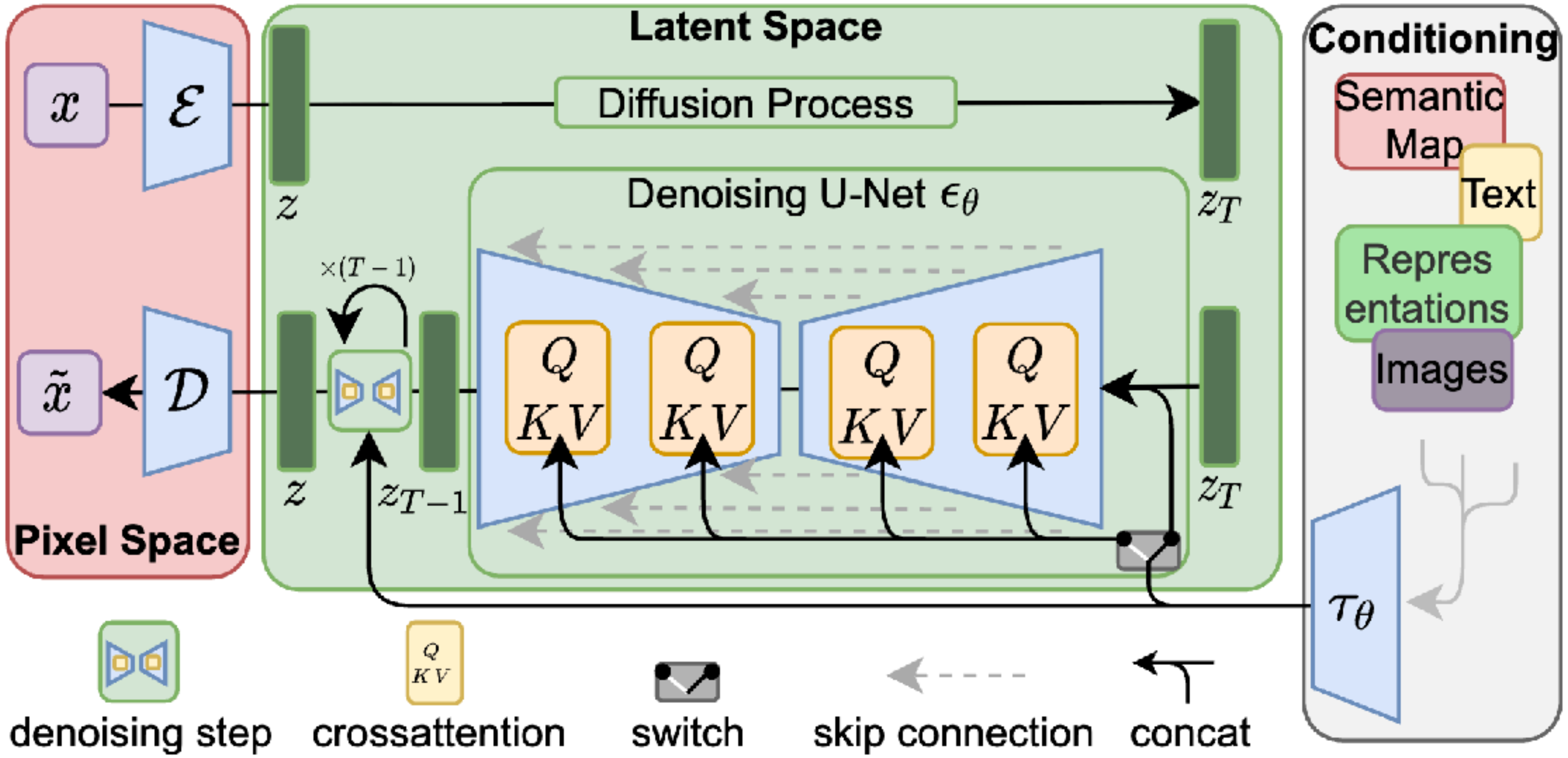
Consistency model



Shortcut model



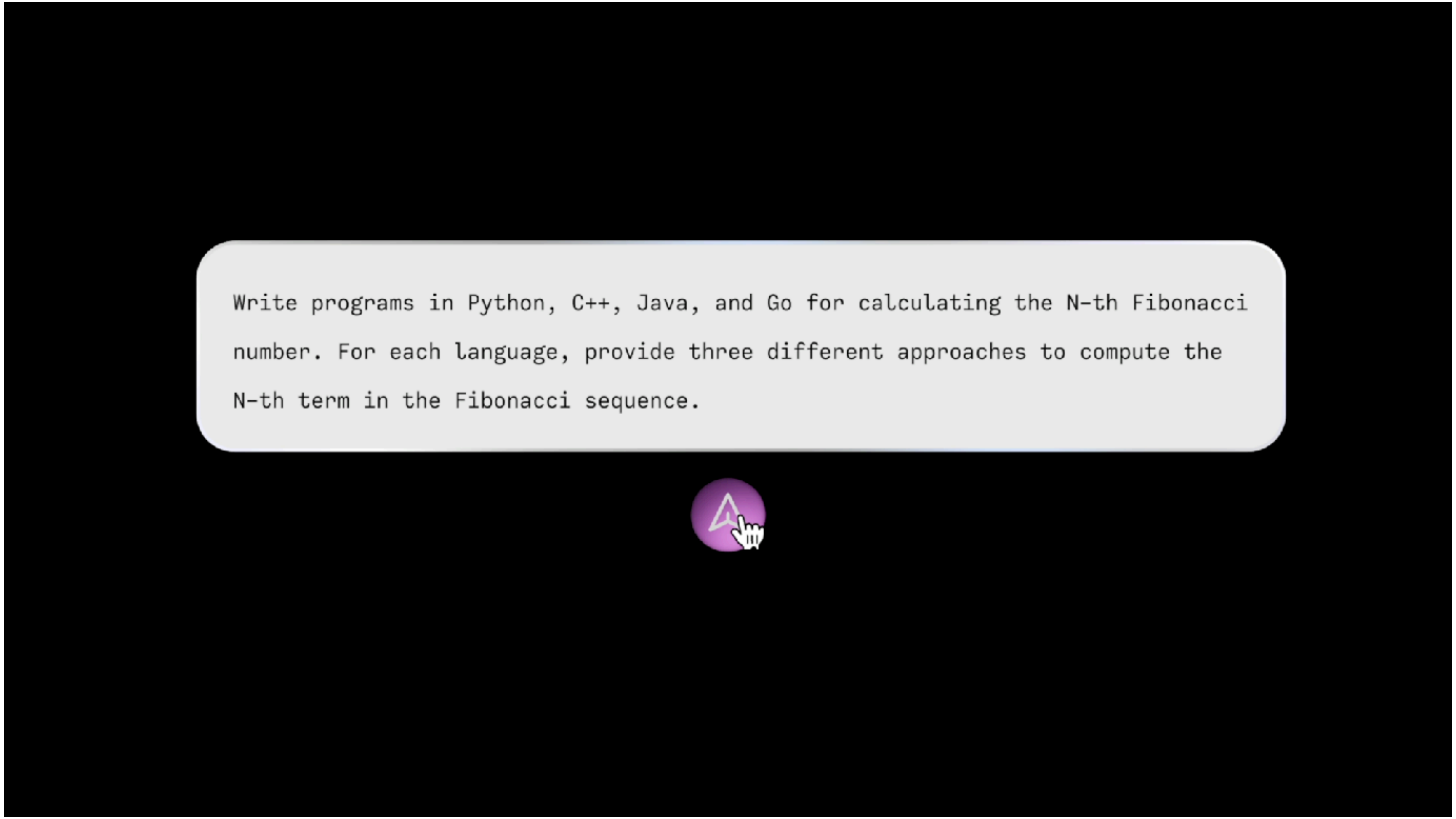
Diffusion on latent space



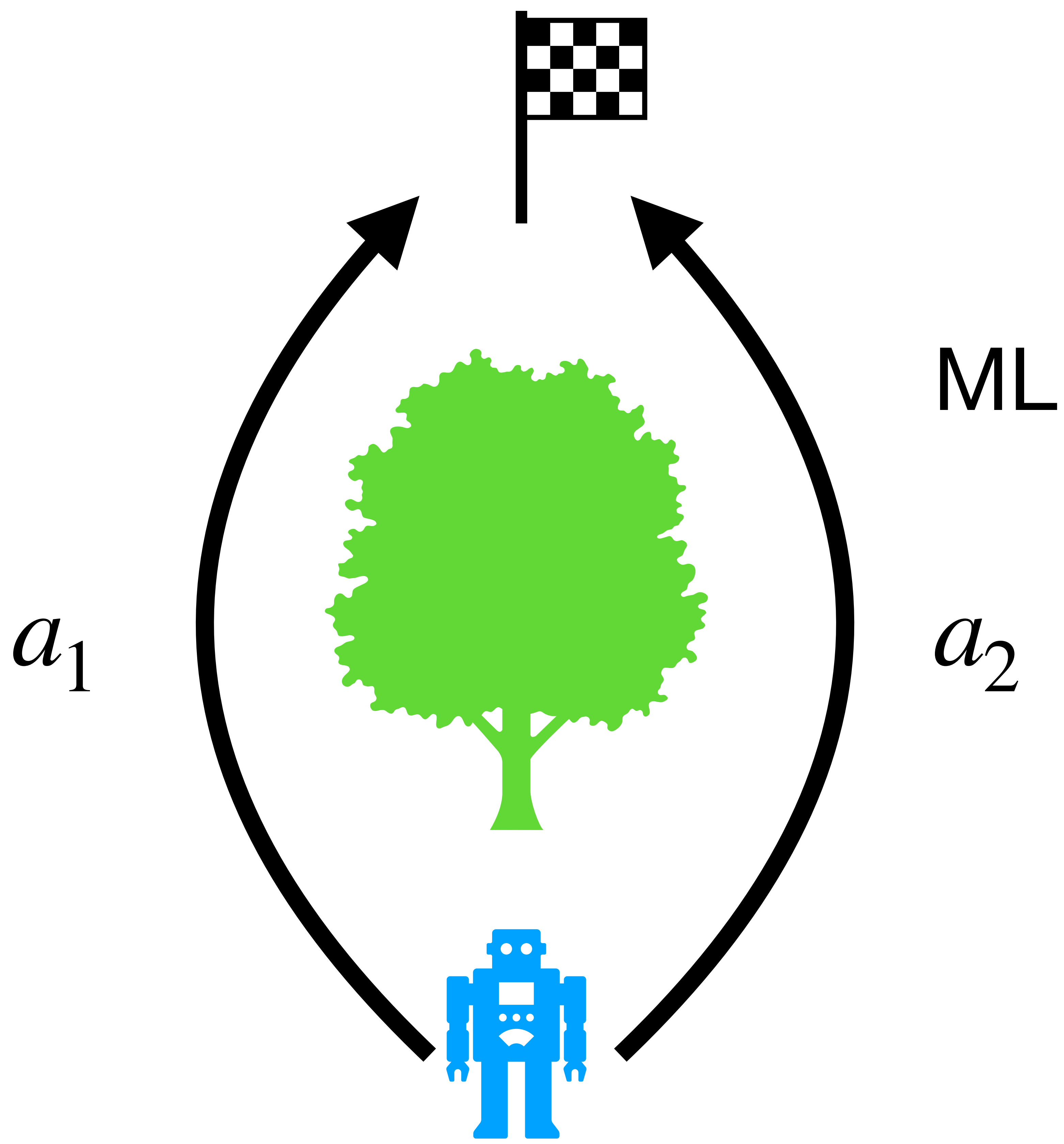
256 x 256 x 3 → 32 x 32 x 4

1. Diffusion Model Basics ✅
2. Efficient Diffusion Sampling ✅
3. Applications Beyond Image Generation

Diffusion model for discrete data

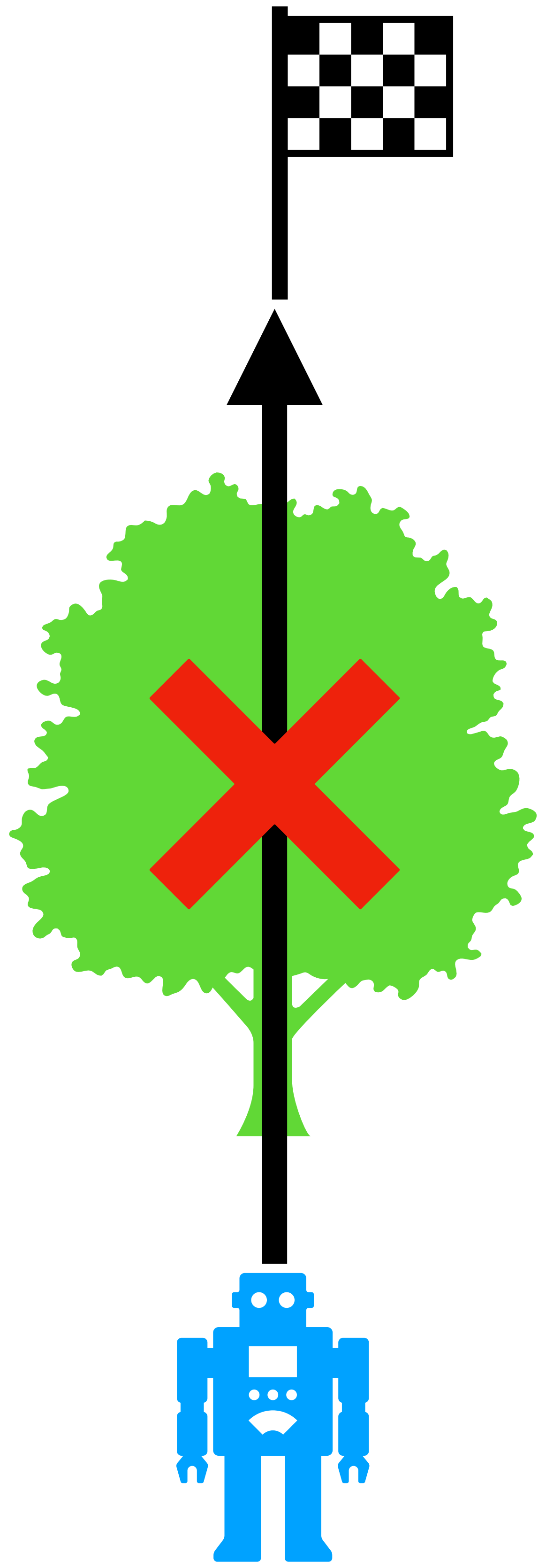


Diffusion policy for robotics

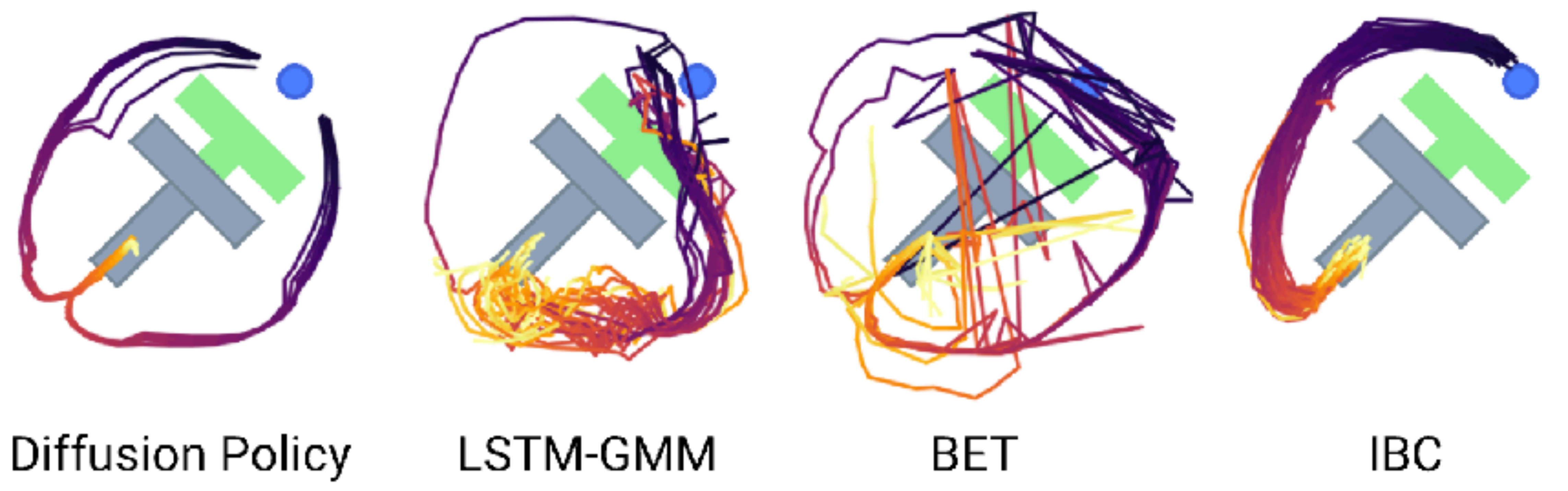


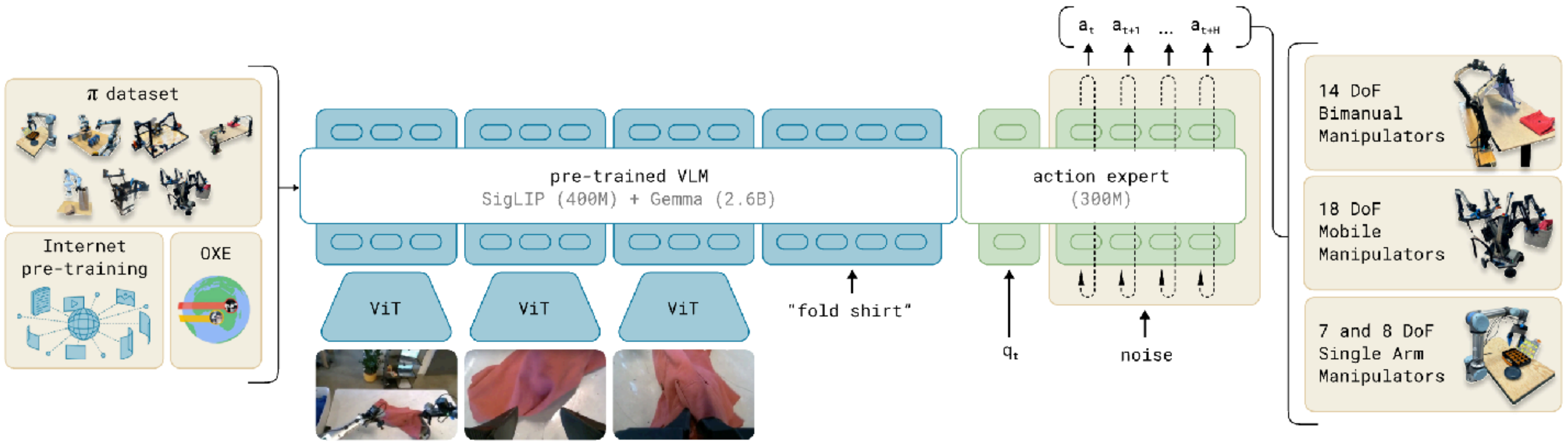
MLE with Gaussian policy on $\{a_1, a_2\}$

$$\hat{a} = \frac{a_1 + a_2}{2}$$



Diffusion policy for robotics





1. Diffusion Model Basics ✅
2. Efficient Diffusion Sampling ✅
3. Applications Beyond Image Generation ✅